MAT1575, Classwork16, Fall2026

5.4 Comparison Tests II (p.426-433)

3. *Theorem*: **Limit Comparison Test**

Let $a_n, b_n \geq 0$ for all $n \geq 1$.

(i) If $\lim_{n\to\infty} \frac{a_n}{b_n} = L \neq 0$, then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge or both diverge.

(ii) If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. ($a_n < b_n$, b_n converges, a_n converges)

(iii) If $\lim_{n\to\infty} \frac{a_n}{b_n} \to \infty$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges. ($0 < b_n < a_n$, b_n diverges, a_n diverges)

4. For each given series, use the limit comparison test to determine whether the series converges or diverges.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^2-1}$ Compare this series with $\sum \frac{1}{n^2}$ (b_n)

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{n^2-1}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^2}{n^2-1} = 1$ (fixed number)

By limit Comparison Test, since $\sum \frac{1}{n^2}$ converg (by p-series), then $\sum_{n=1}^{\infty} \frac{1}{n^2-1}$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}+1}$ (a_n) Compare this series with $\sum \frac{1}{\sqrt{n}}$ (b_n) (diverges, by p-series)

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{\sqrt{n}+1}}{\frac{1}{\sqrt{n}}} = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n}+1} = 1$

Then, by limit Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}+1}$ diverges.

(c) $\sum_{n=1}^{\infty} \frac{2^n+1}{3^n}$ Compare this series with $\sum_{n=1}^{\infty} \frac{2^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ (Converges since $r = \frac{2}{3} < 1$)

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{2^n+1}{3^n}}{\frac{2^n}{3^n}} = \lim_{n\to\infty} \frac{2^n+1}{3^n} \cdot \frac{3^n}{2^n} = \lim_{n\to\infty} \frac{2^n+1}{2^n} = \lim_{n\to\infty} 1+\frac{1}{2^n} = 1$

Then, by limit Comparison test, $\sum_{n=1}^{\infty} \frac{2^n+1}{3^n}$ converges

(d) $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$

5.5 Alternating Series (p.426-433)

1. *Definition*: **Alternating Series**.

Any series whose terms alternate between is called an alternating series and it can be written in the form with $b_n > 0$ for all positive integers n:

or

2. *Examples.*

(a) $\sum_{n}^{\infty}\left(-\frac{1}{2}\right)^{n}$ is

(b) $\sum_{n}^{\infty}\frac{(-1)^{n+1}}{n}$ is

3. *Theorem*: **Alternating Series Test**

An alternating series of the form $\sum_{n=1}^{\infty}(-1)^{n+1}b_n$ or $\sum_{n=1}^{\infty}(-1)^{n}\,b_n$ converges if $(i)\ 0 < b_{n+1} \leq b_n$ for all n ≥ 1 and $(ii)\ \lim_{n\to\infty} b_n = 0$

4. For each given alternating series, determine whether the series converges or diverges.

(a) $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}$

(b) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{n+1}$

5. *Definition*: **Absolute Convergence and Conditional Convergence**

A series $\sum_{n=1}^{\infty}a_n$ exhibits if

A series $\sum_{n=1}^{\infty}a_n$ exhibits if

6. Example of Conditional Convergence:

7. What is the difference between absolute and conditional convergence?

8. For each given series, determine whether the series converges absolutely, converges conditionally, or diverges.

(a) $\sum_{n}^{\infty}\frac{(-1)^{n+1}}{3n+1}$

(b) $\sum_{n}^{\infty}\frac{\cos(n)}{n^2}$